

# Efficient In-Context Learning for 3D Medical Image Segmentation

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by Tidiane Camaret Ndir



# Master's Thesis

## **Efficient In-Context Learning for 3D Medical Image Segmentation**

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# Declaration

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I also hereby declare that my thesis has not been prepared for another examination or assignment, either in its entirety or excerpts thereof.

Freiburg, September 28, 2026

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Tidiane Camaret Ndir

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## Abstract

# **Zusammenfassung**

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# 1 Motivation

In-context learning (ICL) recasts segmentation as few-shot matching: given a target scan and a small set of (image, mask) context pairs illustrating the structure of interest, a model predicts a mask for the target with no task-specific fine-tuning. This removes the need to retrain for every new anatomical class – valuable for structures that are rare, newly defined, or annotated at only a handful of sites.

Most ICL segmentation models were designed and validated in 2D. Moving to native 3D volumes is attractive – CT/MRI are inherently volumetric, and slice-wise processing discards through-plane context needed to resolve thin or elongated structures – but it sharpens an already difficult resolution/compute trade-off: dense cross-attention between a target volume and its context volumes scales with the cube of side length, so a design affordable at  $64^3$  becomes impractical at  $256^3$ .

This trade-off is not one problem but three, and this thesis addresses each with its own mechanism:

1. **How should image and label evidence be combined?** Concatenating image and mask channels before a shared encoder, or fusing them once through a single operation, commits to one fusion point before most of the network’s capacity is applied. section 3 proposes *dual attention*, keeping the two streams separate and letting them exchange information repeatedly over a modality axis and a position axis.
2. **Where should compute be spent?** Re-scanning an entire volume with a sliding window at every processing step wastes compute on regions already known not to contain the structure of interest. section 3 proposes a *coarse-to-fine cascade* that restricts fine processing to the region a coarse pass has already localized, and passes each level’s prediction to the next level as a query prior.
3. **Where does training signal come from beyond limited annotation?** Real (image, mask) pairs are limited to the classes and appearance distribution covered by the training cohort. section 3 proposes a *synthetic task generator* that repaints real anatomical geometry with novel, controllable appearance, producing label-perfect training tasks beyond the annotated cohort.

**Contributions.**

- A dual attention mechanism for in-context 3D segmentation, contrasted against early-concatenation (Medverse [8]) and one-shot fusion (Iris [6]) baselines.
- A coarse-to-fine cascade that seeds each level with the previous level's prediction and restricts fine processing to the region it localizes, compared against the full-volume autoregressive inference used by Medverse.
- A GMM-based synthetic task generator with a texture-noise variant that better matches real tissue's spatial autocorrelation, evaluated against real-only and i.i.d.-noise training.
- Evaluation on TotalSegmentator CT and a range of out-of-distribution cohorts and modalities, reporting where each mechanism helps and where it does not.

**Outline.** section 2 introduces in-context segmentation, the volumetric encoder and attention machinery the method builds on, and synthetic-data background, then positions UniverSeg, Iris, and Medverse against the three axes above. section 3 details the model. section 4 evaluates each axis in turn. section 5 synthesizes what the evidence supports and what remains open, and section 6 summarizes.

## 2 Fundamentals

### 2.1 In-Context Segmentation

Given a target image  $I^t$  and a context set  $\{(I_k^c, L_k^c)\}_{k=1}^K$  of image/mask pairs for the same class, in-context segmentation predicts a mask  $\hat{L}^t$  for the target in a single forward pass, with no gradient update and no class-specific parameters. This sits between two more familiar paradigms: closed-set supervised segmentation, which fixes a label set at training time and must be retrained (or fine-tuned) to add a class; and interactive segmentation, where a human supplies clicks or boxes on the target image itself, per image, at inference time. In-context segmentation instead supplies the “prompt” as a handful of *other*, already-labeled examples of the same class – closer to how a radiologist might be shown a few example cases of a rare finding and asked to find it in a new scan. This is valuable in medical imaging specifically because annotation requires domain expertise, new structures and pathologies are defined faster than segmentation datasets can be built for them, and per-site protocol differences limit how far a closed-set model trained elsewhere transfers.

### 2.2 Volumetric Encoders and Attention

**Convolutional volumetric backbones.** 3D U-Net [4] and its self-configuring descendant nnU-Net [9] encode a volume through a downsampling encoder into a feature pyramid, then decode back to full resolution via a mirrored upsampling path with skip connections carrying fine detail past the bottleneck. Deeper encoder stages have coarser spatial extent and more channels; shallower stages retain spatial detail at low channel count. This split – a coarse, semantically rich bottleneck feeding a computationally light attention stack, and fine, spatially precise early stages reserved for the decoder – is exactly what the coarse/fine tap split in subsection 3.1 reuses. UNETR [7] is a prior example of combining a convolutional decoder with transformer attention for 3D segmentation.

**Attention.** Scaled dot-product attention [14] computes, for each query token, a weighted average of value vectors, with weights given by the similarity between the query and each key:

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V. \quad (1)$$

Multi-head attention runs several such operations in parallel subspaces and concatenates the results, letting different heads attend to different relationships in the same input.

**Position encoding.** Attention itself is permutation-invariant, so position must be injected separately. Rotary position embedding (RoPE) [12] does this by rotating each query/key vector by an angle proportional to its position before the dot product, so the result depends only on the *relative* position between the two tokens rather than their absolute indices. In 3D, this is applied axially: the head dimension is split into three chunks, one per spatial axis, each rotated by an angle proportional to that axis’s physical coordinate (scaled by voxel spacing) – letting attention between two cells at different cascade resolutions still reflect their true physical distance, not just their grid index.

## 2.3 Synthetic Training Data

Domain randomization trains a model on procedurally varied synthetic data so it does not overfit to the narrow appearance distribution of a single acquisition protocol or training cohort. SynthSeg [1] demonstrated this at scale for medical segmentation: training pairs are generated by sampling a random Gaussian-mixture appearance conditioned only on a label map, so the resulting segmentation network never sees a real image of any specific contrast during training yet generalizes across MRI contrasts and resolutions. GIN/IPA [10] takes a complementary, appearance-only approach to single-source domain generalization: a random nonlinear intensity warp (GIN), optionally blended across spatially distinct regions of the same volume (IPA) to break spurious global appearance correlations, achieving strong generalization from randomizing appearance alone, without touching shape or geometry. Both results motivate treating appearance randomization as a distinct lever from geometric augmentation – a question the synthetic task generator (subsection 3.4) revisits directly by ablating *how* painted appearance is randomized (i.i.d. vs. spatially correlated texture noise) while holding geometry fixed.

## 2.4 Related Work

Three prior systems attempt in-context medical image segmentation, each making a different choice on the three axes this thesis organizes around.

**UniverSeg [2]** popularized the approach with a CrossBlock architecture that densely cross-attends target and context features at full resolution – no coarse-to-fine structure, and no synthetic-task training.

**Iris [6]** encodes each context pair once into a compact task embedding, using pixel shuffle operations in its context encoder. The target interacts with the context only through this embedding, in the mask decoder, and inference is a single pass with no refinement across scales.

**Table 1:** Positioning against prior in-context segmentation models on the three axes this thesis organizes around.

|             | Fusion                    | Coarse-to-fine                       | Synthetic tasks       |
|-------------|---------------------------|--------------------------------------|-----------------------|
| UniverSeg   | dense cross-attn.         | no                                   | no                    |
| Iris        | one-shot pixel shuffle    | no                                   | no                    |
| Medverse    | dual U-Net cross-attn.    | full-volume re-scan                  | no                    |
| <b>Ours</b> | dual attention (repeated) | region-restricted, prediction-seeded | GMM repaint + texture |

**Medverse [8]** uses separate 3D U-Nets for the target and the context and exchanges features between them at every stage: target-to-context in the encoder, context-to-target in the decoder. The exchange uses a Blockwise Cross-Attention Module (BAM), which mean-pools queries and keys over a  $4^3$  block grid before attending, while values stay at full resolution.

**Hierarchical, coarse-to-fine approaches.** Deep Neural Patchworks [11] stacks levels of increasing resolution for large-volume segmentation, but targets single-model supervised segmentation rather than in-context, few-shot prediction – an architectural precedent for coarse-to-fine cascades outside the ICL setting. Medverse applies a similar idea within the ICL setting through next-scale autoregressive segmentation, inspired by next-scale image generation [13]. The first step predicts the whole volume at low resolution. Each later step doubles the resolution and processes the entire volume with a sliding window, feeding the previous step’s prediction into a third U-Net branch that shares weights with the context branch.

## 3 Methodology

### 3.1 Problem Setup and Backbone

Given a target volume  $I^t$  and a context set  $\{(I_k^c, L_k^c)\}_{k=1}^K$  of image/mask pairs for the same anatomical class as the (unknown) target mask, the model predicts a target mask  $\hat{L}^t$ . Prediction proceeds through a coarse-to-fine cascade of  $M$  resolution levels with decreasing voxel spacing  $r_1 > r_2 > \dots > r_M$ ; each level produces a full mask that seeds the next, finer level (subsection 3.3). Figure 1 shows our architecture (Medverse’s, for comparison, is shown in Figure 5 in the appendix).

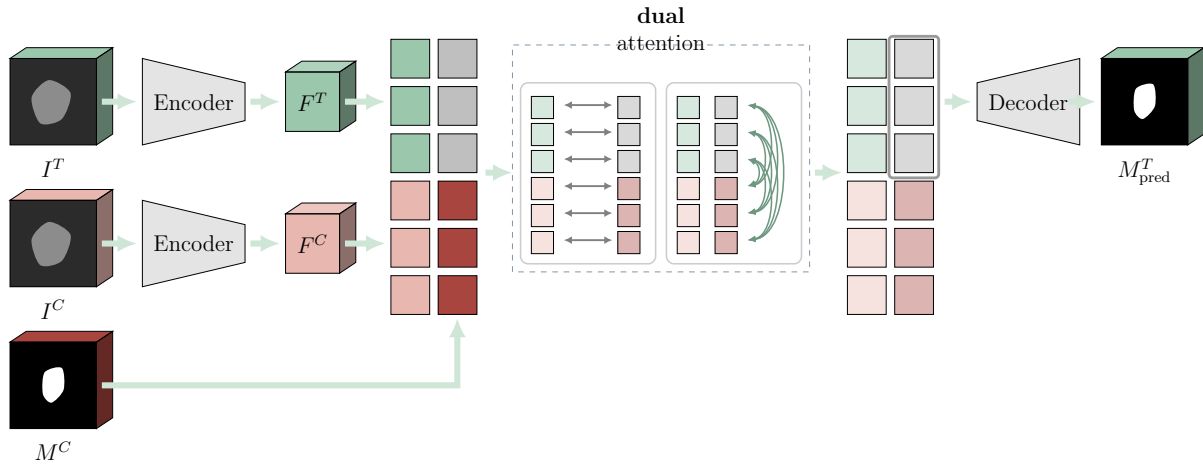
**Shared encoder.** All  $K + 1$  volumes of a task are passed once through a shared encoder, producing a feature pyramid per volume. Two disjoint subsets of its stages are read out: *coarse* stages, resampled to a common  $R \times R \times R$  token grid ( $N = R^3$  cells per volume) and consumed by the attention stack below; and *fine* stages, kept at native resolution and reserved for the decoder’s skip connections (subsection 3.2). This split keeps the attention stack on a small, fixed-size token grid while still allowing a full-resolution decoded mask.

**Tokenization.** Each cell of every volume yields two tokens sharing one row – an image token  $z^{\text{img}}$  and a label token  $z^{\text{lbl}}$  – assembled into a single sequence together with a small number of learned *register tokens*, shared across all volumes, that give the attention stack fixed-size read/write capacity not tied to any single spatial cell. Context rows carry the ground-truth mask in their label token; the target’s does not, since that is exactly what is being predicted – it instead starts from a neutral placeholder or, at cascade levels beyond the first, the previous level’s own prediction (subsection 3.3).

### 3.2 Dual Attention: Modality and Position Axes

How a model combines image and label evidence, and how context and target interact, are two related design choices this section addresses together. Medverse [8] concatenates image and label channels before a shared U-Net; context and target then interact through a Blockwise Cross-Attention Module (BAM) placed at two points in that network (once encoder-side, once decoder-side), which pools its queries and keys to block resolution before attending – trading positional precision for a cheaper attention cost – while its values stay at full resolution. Iris [6] fuses image and label features once, through a pixel shuffle–unshuffle operation, then lets context inform the target through a single bidirectional cross-attention step against a compressed, per-class task embedding computed once and reused across every query. Both commit every design choice – fusion point and context–target interaction alike – to a single, early

Draft diagram, ported from the conference draft; finalize labels/scale once the architecture is settled.



**Figure 1:** Architecture overview: separate image/label streams + dual attention.

step, computed once. Figure 2 contrasts Medverse’s dual, separately-weighted per-row encoders against the shared-encoder family both our baseline and final design belong to.

We instead keep image tokens  $z^{\text{img}}$  and label tokens  $z^{\text{lbl}}$  (subsection 3.1) as separate streams throughout the network and let them exchange information repeatedly, through *dual attention* – following the general pattern of Fu et al. [5], who fuse a position-wise and a channel-wise attention branch at every layer of a scene-segmentation network – with two attention axes applied at every layer, on the full, uncompressed token grid:

- **Modality axis:** at each cell, the image and label token attend to each other, letting label evidence sharpen the image representation and vice versa, independently per cell.
- **Position axis:** every cell of every volume – context, target, and the register tokens – attends over the entire token sequence at once, using 3D axial rotary position embeddings so attention is a function of physical distance between cells rather than raw grid index. This single joint attention performs within-volume self-attention and cross-context matching together, since tokens from every volume already share one sequence.

Figure 3 contrasts this dual attention against a single-fusion-step baseline within the same shared-encoder family.

Only the target’s post-attention *image* token is read out for decoding; the label token is discarded after the last layer, since it started from a placeholder rather than real

Draft diagram, ported from the conference draft; finalize once the layer factorization is settled.

Draft diagram, ported from the conference draft; fi-

image evidence for the target row. The decoded image tokens are projected back to the coarse token grid and progressively fused with the encoder’s fine-resolution skip features (subsection 3.1) via upsampling and convolution, one step per fine stage, into the final full-resolution mask logits.

### 3.3 Coarse-to-Fine Cascade

The cascade connects consecutive resolution levels through two mechanisms: a query prior and region-restricted fine processing.

**Query prior.** Medverse passes the previous step’s prediction, together with the target image, through a third U-Net branch that shares weights with its context branch; during training, this input is a downsampled ground-truth mask standing in for a coarse prediction. Iris has no multi-step inference and uses no prior. At level  $\ell > 1$ , we instead initialize the target’s label token directly from level  $\ell-1$ ’s own prediction:

$$z_{\ell,0}^{\text{lbl}} = \text{LblEmbed}(\text{Warp}_{\ell-1 \rightarrow \ell}(\hat{L}_{\ell-1}^t)), \quad (2)$$

where  $\text{Warp}$  resamples level  $\ell-1$ ’s prediction onto level  $\ell$ ’s crop grid, correcting for the two levels’ different centers, field of view, and any augmentation applied independently to each. This makes the previous level’s belief part of the same token stream processed by dual attention alongside the image (subsection 3.2), rather than a separate side channel. The previous level’s prediction is detached before warping, so no gradient flows between levels through the prior. Figure 4 shows this flow across two levels.

**Region-restricted fine processing.** Medverse processes the entire volume with a sliding window at every step after the first. Once a coarse level has localized the structure of interest, we instead restrict the sliding window at the next, finer level to the coarse prediction’s foreground region, thresholded and padded by a fixed margin. Compute at fine levels then scales with the size of the structure rather than the size of the input volume, at the cost of depending on the coarse level’s localization being approximately correct.

### 3.4 Synthetic Task Generation

A fraction  $p_{\text{synth}}$  of training tasks are drawn from a synthetic generator instead of real context/target subjects, broadening training beyond the anatomical classes and appearance distribution the real cohort covers.

Draft diagram, ported from the conference draft; finalize once the number of levels  $M$  is settled.

**Procedural shape generator.** Each synthetic task samples a binary target mask from one of seven procedural shape families (blob, splatter, disk, cylinder, scatter\_field, vessel, torus) and paints it onto a canvas image. Paint intensity is host-anchored – drawn relative to the canvas region’s own local mean/std rather than an absolute value – and textured with multi-octave, spatially correlated noise (matching real tissue’s autocorrelation, unlike flat i.i.d. paint), with an optional core/rim intensity blend for added heterogeneity. *scatter\_field* specifically targets lesion *multiplicity* (many small, spatially scattered components), a pattern most anatomical structures in the real training cohort do not exhibit.

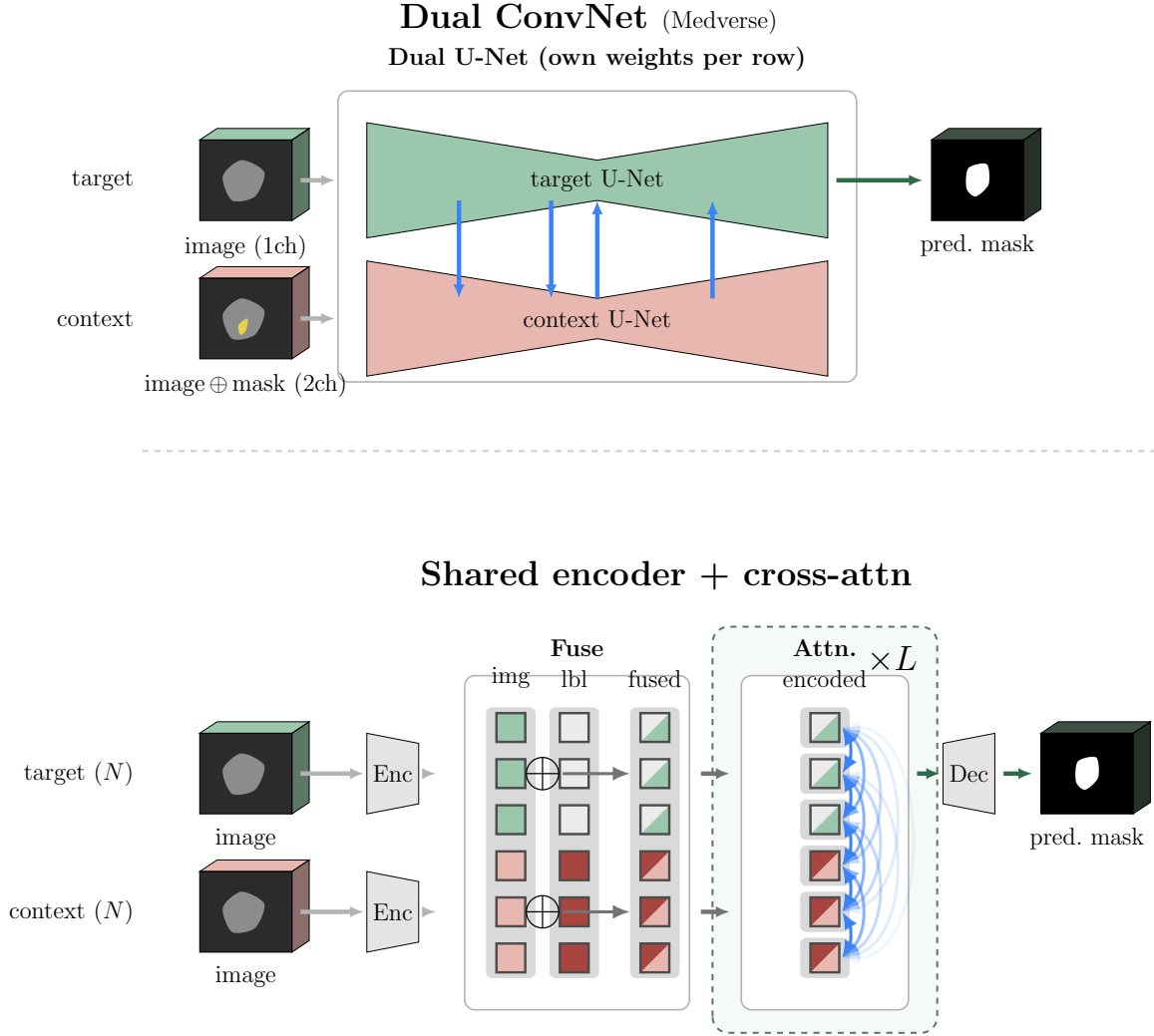
**Canvas: synthetic bank vs. real subject.** Two canvas sources are available. The *synthetic-bank* canvas draws a candidate mask/image region from a precomputed bank of segmented real scans and paints the shape into it. The *real-host* canvas instead paints the same procedural shape directly onto a real TotalSegmentator training case (CT or MRI, any organ) , bypassing the bank entirely: every volume in a task shares one real subject as its base image, differing only in the painted region.

### 3.5 Training Objective

Each cascade level is supervised independently at its own resolution with a combined binary cross-entropy and soft Dice loss, summed over the  $M$  levels:

$$\mathcal{L} = \sum_{\ell=1}^M \lambda_{\ell} (\mathcal{L}_{\text{BCE}}^{\ell} + \mathcal{L}_{\text{Dice}}^{\ell}) . \quad (3)$$

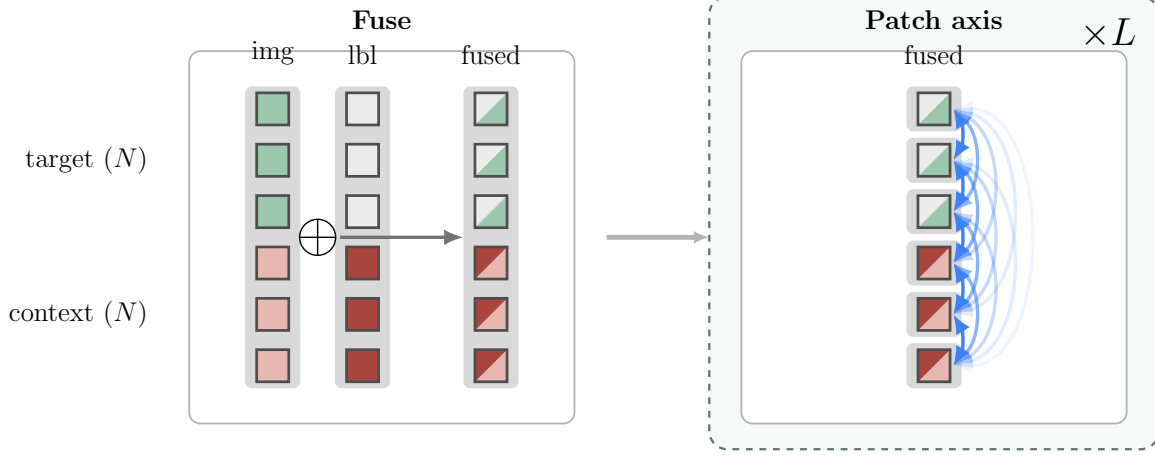
Level weights  $\lambda_{\ell}$  are uniform ( $\lambda_{\ell}=1$ ). Because the query prior is detached (subsection 3.3), each level’s loss does not back-propagate through the coarser level’s prediction.



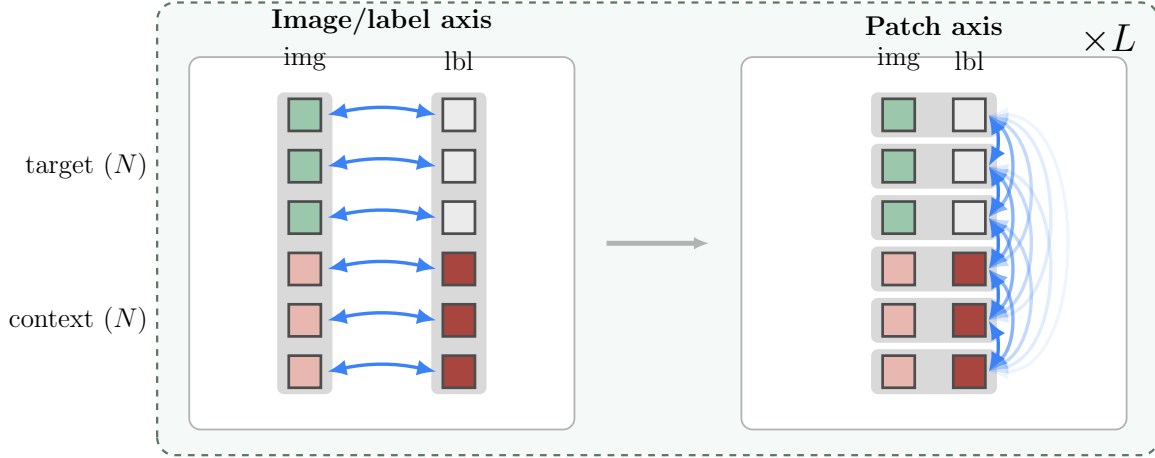
**Figure 2:** Two families of context/target coupling, shown for a single context volume ( $K=1$ ) and 3 of its  $N$  tokens per stream. Top: Medverse channel-concatenates image and mask into one volume (context only – target gets no mask channel at all) and feeds each row through its own, separately-weighted U-Net, coupled by cross-attention at several stages: context queries target’s encoder while narrowing, target queries context’s decoder while widening (blue arrows). Bottom: the family both our baseline and final design belong to instead passes only the image through one shared encoder (“Enc”), run once; label evidence – a placeholder prior for target, the real mask for context – re-enters at a Fuse step ( $\oplus$ ) before the encoded token attends across cells and contexts for  $L$  layers (“Attn.”) and a shared decoder (“Dec”) produces the mask. Figure 3 details the two ways we combine image and label evidence within this shared-encoder family.

## Shared encoder + cross-attn

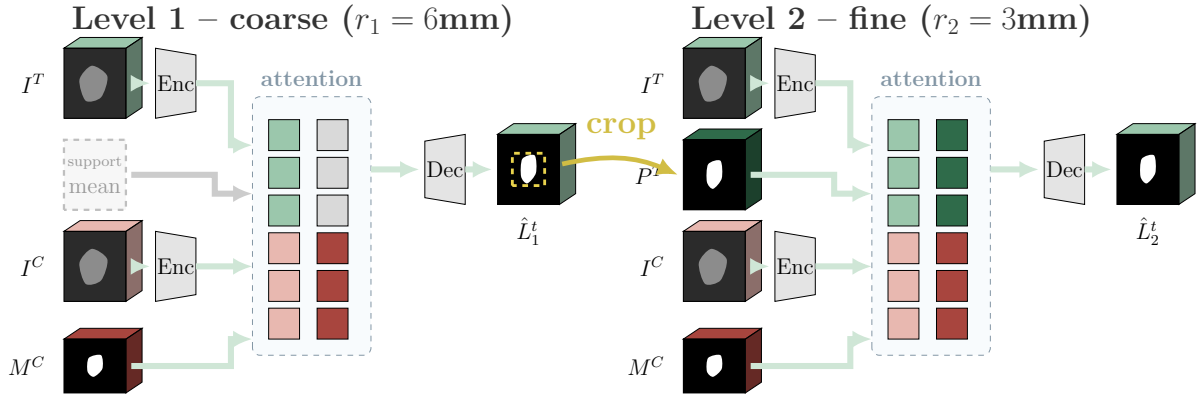
Baseline: early fusion



Ours: dual attn.



**Figure 3:** Two variants within the shared-encoder family (Figure 2), shown for a single context volume ( $K=1$  for clarity) and 3 of its  $N$  spatial tokens per stream. Top: early fusion (our baseline) – image and label tokens are merged once ( $\oplus$ ), then the fused token grid attends across cells and contexts for  $L$  layers (the position axis). Bottom: dual attention (ours) – image and label tokens instead attend to each other directly at every layer (the modality axis, blue double arrows), alongside the same cross-cell/cross-context attention (the position axis), repeated together for  $L$  layers. In both, the target row (green image tokens) cross-attends read-only into context (pink tokens); its label-stream input is the query prior (subsection 3.3) rather than ground truth, which is why its label token is gray rather than a role color.



**Figure 4:** Query image prior across two cascade levels, with the full model input/output flow. At level 1 the target has no real prior yet (dashed support-mean placeholder); at level 2 the target's second lane is a real crop of level 1's own prediction  $\hat{L}_1^t$  – the same lane the context mask always occupies.

## 4 Results

### 4.1 Experimental Setup

**Training data and model scope.** All models are trained on the training cases of TotalSegmentator, restricted to its training classes; the remaining classes form the unseen split. Each results section uses its own training setup. The fusion comparison (subsection 4.2) trains all three arms on the CT subset only, at a single resolution level. The cascade experiments (subsection 4.3) train the reference configuration (dual attention with query-prior initialization) on both CT and MRI. The synthetic-task experiments (subsection 4.4) train an extended configuration, which adds cross-level register carry, a pooling token, and synthetic texture noise, also on CT and MRI. Numbers from different sections are therefore not directly comparable. In particular, TotalSeg MRI is a held-out modality only in the fusion section.

**Baselines.** Medverse [8] (released weights, and – where noted – finetuned on the same TotalSegmentator split) is the only baseline evaluated empirically here.

**Metrics.** Accuracy is reported as Dice (macro, per-class then averaged, unless stated per-sample) and Normalized Surface Dice (NSD). Compute is reported as GFLOPs, wall-clock latency, and peak VRAM where available; VRAM comes from a standalone  $B=1, K=1$  inference benchmark rather than the evaluation loop itself, so it should not be read as representative of a real  $K>1$  eval run.

**Datasets.** Table 2 lists every evaluation source used in this thesis, grouped by relationship to the training vocabulary. TotalSegmentator (the in-distribution source) and FLARE22 both reuse the training vocabulary [15], splitting into a genuine seen/unseen pair; the rest introduce a target, class, or scanner modality absent from training entirely.

### 4.2 Image–Label Fusion

Two comparisons follow, each isolating one design choice. The first asks whether moving from Medverse’s dual U-Net and cross-attention to a shared encoder with attention already helps, before anything else about this thesis’s architecture is considered; the second asks whether keeping image and label evidence as separate streams – dual attention – helps further, within that same encoder-and-attention family.

**Setup.** Table 1 classifies Medverse’s fusion strategy as dual U-Net cross-attention: target and context are encoded by two separate U-Nets and fused via cross-attention between the two streams at every stage, rather than through a single shared encoder.

Resolve before submission: does the thesis report a single “Ours” configuration, or keep this split?

No Iris baseline exists yet; the comparison in subsection 2.4 remains architectural, not empirical.

Add

**Table 2:** Every evaluation source, grouped by relationship to the training vocabulary.  $n$  is subjects;  $n_{\text{cls}}$  is evaluated classes.

| Source                                           | Modality   | Target                                            | $n$ | $n_{\text{cls}}$ |
|--------------------------------------------------|------------|---------------------------------------------------|-----|------------------|
| <i>In-vocabulary (seen/unseen split)</i>         |            |                                                   |     |                  |
| TotalSegmentator                                 | CT         | 117 organs (61 seen / 56 unseen)                  | 89  | 117              |
| FLARE22                                          | CT         | abdominal organs (11 seen / 2 unseen)             | 50  | 13               |
| <i>Novel vocabulary (unseen by construction)</i> |            |                                                   |     |                  |
| ISLES22                                          | MRI DWI    | stroke lesion                                     | 250 | 1                |
| Shifts-MS                                        | MRI FLAIR  | MS lesion                                         | 46  | 1                |
| MSD Hippocampus                                  | MRI T1     | hippocampus                                       | 260 | 2                |
| MSD Prostate                                     | MRI T2+ADC | prostate zone                                     | 64  | 2                |
| ATLAS v2.0*                                      | MRI T1     | chronic stroke lesion                             | 654 | 1                |
| GNC_705                                          | MRI Dixon  | kidney lesion                                     | 610 | 13               |
| NasalSeg                                         | CT         | nasal/sinus cavities                              | 107 | 5                |
| <i>Non-object</i>                                |            |                                                   |     |                  |
| HU_LWK1                                          | CT         | L1 vertebra ROI                                   | 36  | 1                |
| <i>Held-out modality</i>                         |            |                                                   |     |                  |
| TotalSeg MRI <sup>†</sup>                        | MRI        | 23 TotalSegmentator classes (?? seen / ?? unseen) | 47  | 23               |

\*Sourced from an unofficial data mirror; provenance not yet independently confirmed.

<sup>†</sup>Held-out modality in the fusion experiments, which train on CT only; in-distribution in the cascade and synthetic-task experiments.

This arm is labeled **Dual U-Net Cross-Attn** throughout, not Medverse: it starts from Medverse’s released weights (the best available starting point for this architecture) but is then trained under the identical recipe as the other two arms for the same  $\sim 250$  epochs – a different thing from the released, unmodified baseline evaluated elsewhere in this thesis (subsection 4.3). The single-axis control isolates the architectural family this comparison targets: it keeps the shared encoder and attention stack common to every architecture in this thesis, but collapses image and label evidence into one token per cell via a pixel shuffle–unshuffle operation *before* the transformer, similar to the pixel shuffle Iris uses when encoding its context, and withholding dual attention until the next step of the staircase.

Its transformer depth is raised from  $l=4$  to  $l=9$  specifically to compute-match the dual-attention arm (removing the modality axis also removes a large share of the model’s compute; parameter count and compute could not be matched simultaneously, leaving parameters 52% higher for the single-axis control). This adjustment is made relative to dual attention only, not to Dual U-Net Cross-Attn, so the Dual U-Net Cross-Attn comparison alone is not compute-matched.

**Table 3:** Training settings shared by Dual U-Net Cross-Attn, the single-axis control, and dual attention (ours). Epoch count is a placeholder pending confirmation of the exact per-arm value.

|                   |                               |
|-------------------|-------------------------------|
| Optimizer         | AdamW                         |
| Learning rate     | $10^{-4}$                     |
| Schedule          | cosine decay, 10-epoch warmup |
| Weight decay      | $10^{-2}$                     |
| Batch size        | 8                             |
| Gradient clipping | 1.0                           |
| Loss              | BCE + Dice                    |
| Epochs            | $\sim 250$                    |

All three arms train under an identical recipe (Table 3), an identical data pipeline, and an identical augmentation recipe, but not an identical starting point: Dual U-Net Cross-Attn keeps its released initialization, while the single-axis and dual-attention arms warm-start only the encoder and decoder from a shared checkpoint, with the entire in-context reasoning core (transformer, image/label embeddings, context/query id, register tokens, cascade projection, pool projection) randomly initialized for both – so neither patchset arm starts from an already-trained transformer. Training uses real anatomy only ( $p_{\text{synth}}=0$ ), on TotalSegmentator CT training cases and classes only. The three arms also did not train to a matched stopping point: Dual U-Net Cross-Attn trained to full convergence, the patchset arms did not.

All three arms are then evaluated with an identical protocol – fixed 3 mm single-level crops, fp32, each model’s own best checkpoint – across every dataset this thesis evaluates (subsection 4.1), not TotalSegmentator alone.

Rows below are grouped by each dataset’s relationship to the training vocabulary, not by how different the source otherwise looks: TotalSegmentator itself (61 seen / 56 unseen classes) and FLARE22 (11 seen / 2 unseen, out of 13 total, subsection 4.1) both reuse the training vocabulary and so split into a genuine seen/unseen pair; MSD Prostate and MSD Hippocampus introduce organs with no vocabulary overlap at all, so every class in that pair is unseen by construction, not by a train/val split; HU\_LWK1’s target is a HU-measurement region, not a segmentable anatomical object, a categorically different kind of “unseen” from a novel organ; TotalSegmentator’s own MRI acquisition isolates a held-out *modality*, since these models see no MRI during training, over classes the vocabulary already contains.

**Results.** Single-axis alone already beats Dual U-Net Cross-Attn on five of the eight rows – both TotalSegmentator splits, both FLARE22 splits, and TotalSeg MRI – while Dual U-Net Cross-Attn wins the three remaining rows: MSD Prostate, MSD Hippocam-

**Table 4:** Dual U-Net Cross-Attn, the single-axis control, and dual attention (ours), across all evaluation datasets (subsection 4.1). Grouped by vocabulary relationship: in-vocabulary seen/unseen, novel-vocabulary (unseen by construction), non-object, held-out modality.

|                            | Dual U-Net Cross-Attn | Encoder + Attention |                       |
|----------------------------|-----------------------|---------------------|-----------------------|
|                            |                       | Single-axis         | Dual attention (ours) |
| TotalSeg CT – seen         | 0.348                 | 0.574               | <b>0.582</b>          |
| TotalSeg CT – unseen       | 0.221                 | 0.408               | <b>0.426</b>          |
| FLARE22 – seen             | 0.532                 | 0.693               | <b>0.709</b>          |
| FLARE22 – unseen           | 0.443                 | <b>0.632</b>        | 0.604                 |
| MSD Prostate (unseen)      | <b>0.377</b>          | 0.205               | 0.247                 |
| MSD Hippocampus (unseen)   | 0.172                 | 0.122               | <b>0.184</b>          |
| HU_LWK1 (non-object)       | <b>0.146</b>          | 0.079               | 0.102                 |
| TotalSeg MRI (modality)    | 0.178                 | 0.232               | <b>0.259</b>          |
| mean of 5 held-out sources | 0.278                 | 0.264               | <b>0.297</b>          |
| $l$                        | –                     | 9                   | 4                     |
| params                     | 71.1M                 | 74.1M               | <b>48.7M</b>          |
| GFLOPs                     | <b>2362.6</b>         | 3973.1              | 3892.5                |
| latency (fp32, per sample) | 73.2 ms               | 73.0 ms             | <b>72.5 ms</b>        |

pus, and HU\_LWK1. Restoring dual attention flips one of those three (MSD Hippocampus, 0.184 vs. 0.172) without touching the other two, bringing the full architecture to six wins out of eight against Dual U-Net Cross-Attn. The one aggregate number in the table – mean of the five held-out sources – shows why that flip matters: single-axis alone is a net regression against Dual U-Net Cross-Attn there (0.264 vs. 0.278), and only once dual attention restores the modality axis does the full architecture overtake Dual U-Net Cross-Attn (0.297). The architectural family change alone, in other words, is not what carries this result; dual attention is. The compute picture is mixed rather than uniformly favorable: dual attention has the fewest parameters and the lowest latency of the three, but not the fewest GFLOPs – Dual U-Net Cross-Attn remains cheapest by that measure despite its lower accuracy, so no arm here is a strict win on both accuracy and compute at once. Two caveats keep this from a single-variable ablation: Dual U-Net Cross-Attn starts from a pretrained initialization rather than the randomly-initialized reasoning core the patchset arms share, and does not receive the same query-conditioning signal during training – so this is best read as a comparison against Dual U-Net Cross-Attn’s strongest available starting point, not a fully symmetric ablation.

**Status.** Neither patchset arm had reached its full training budget at the point these numbers were read, while Dual U-Net Cross-Attn had already converged – so Table 4 may understate Dual U-Net Cross-Attn’s eventual ceiling more than either patchset arm’s. All three arms are N=1 seed, and several rows rest on very few classes (FLARE22’s

Re-read  
once  
Dual  
U-Net  
Cross-  
Attn  
receives  
the  
same

**Table 5:** Predicted-prior checkpoint vs. Medverse (native autoregressive, depth-matched), macro Dice and mean per-sample latency. Levels/spacing is our own cascade ladder for that dataset; Medverse’s AR depth is matched to the same level count.

| Dataset         | Levels | Spacing (mm)   | Dice         |              | Latency      |          |
|-----------------|--------|----------------|--------------|--------------|--------------|----------|
|                 |        |                | Ours         | Medverse     | Ours         | Medverse |
| TotalSeg CT     | 3      | [6, 3, 1.5]    | <b>0.586</b> | 0.103        | 278ms        | 4978ms   |
| FLARE22         | 3      | [6, 3, 2.5]    | <b>0.751</b> | 0.359        | 311ms        | 4894ms   |
| MSD Prostate    | 3      | [3, 1.5, 0.75] | <b>0.375</b> | 0.357        | 279ms        | 5006ms   |
| MSD Hippocampus | 1      | [0.5]          | 0.303        | <b>0.686</b> | 99ms         | 50ms     |
| HU_LWK1         | 3      | [6, 3, 1]      | <b>0.152</b> | 0.020        | 439ms        | 5336ms   |
| TotalSeg MRI    | 3      | [6, 3, 1.5]    | <b>0.506</b> | 0.038        | 306ms        | 4842ms   |
| NasalSeg        | 2      | [1.5, 0.8]     | 0.547        | <b>0.737</b> | 161ms        | 1886ms   |
| mean            |        |                | –            | –            | <b>268ms</b> | 3856ms   |

unseen split, both MSD sources, HU\_LWK1) – directional evidence, not settled effect sizes.

### 4.3 Coarse-to-Fine Cascade

This section asks whether the coarse-to-fine cascade introduced in subsection 3.3 helps in-distribution at all; how the resulting design compares to Medverse’s own coarse-to-fine mechanism, on accuracy and compute; and which of the cascade’s own mechanisms are responsible for the in-distribution gain.

**Coarse-to-fine cascading improves accuracy in-distribution.** A spacing sweep from 4mm to 1.5mm crop resolution (117 classes, 3897 samples) raises macro Dice from **0.361 to 0.538** (+0.177). The locator’s containment of the target structure is already high at the coarse level (0.922 mean, vs. an oracle ceiling of 0.970 – a 0.048 gap) and does not predict which classes gain the most at the fine level: once a class is roughly localized, the remaining error is dominated by segmentation quality at the fine crop, not by further localization failure.

**Ours vs. Medverse, seven datasets, accuracy and compute.** “Ours” here is the predicted-prior checkpoint (the best-performing configuration of the query-prior ablation below); “Medverse” is released weights, native autoregressive inference – its own coarse-to-fine mechanism – with its refinement depth level-matched to our own cascade ladder per source – the fairer comparison, even though matching that depth actually *hurts* Medverse’s own accuracy. Four held-out sources (ISLES22, Shifts-MS, ATLAS v2.0, GNC\_705) are excluded – both models score too low there to be informative. Table 5 reports macro Dice and mean per-sample latency for both.

Re-read once all three arms reach their full training budget; add  $\geq 2$  seeds before citing any effect size as settled.

TotalSeg CT’s Medverse

**Table 6:** Query-prior source ablation, macro Dice, across all validation datasets. TotalSeg CT also reports NSD (0.576 / **0.631**).

| Dataset         | GT-prior (perturbed) | Predicted prior (real prev. pred) |
|-----------------|----------------------|-----------------------------------|
| TotalSeg CT     | 0.543                | <b>0.586</b>                      |
| FLARE22         | 0.731                | <b>0.751</b>                      |
| MSD Prostate    | 0.310                | <b>0.375</b>                      |
| MSD Hippocampus | <b>0.319</b>         | 0.303                             |
| HU_LWK1         | 0.099                | <b>0.152</b>                      |
| TotalSeg MRI    | 0.449                | <b>0.506</b>                      |
| NasalSeg        | 0.399                | <b>0.547</b>                      |

Ours wins 5/7 (TotalSeg CT, FLARE22, MSD Prostate, HU\_LWK1, TotalSeg MRI); Medverse wins 2/7 (MSD Hippocampus, NasalSeg) – both cases where the target is a single, large, well-defined structure that suits Medverse’s full-FOV multi-resolution pyramid. Compute favors Ours by roughly  $14\times$  on mean latency – and this is *after* level-matching Medverse’s own depth to ours, which costs Medverse accuracy too, so an unmatched Medverse would lose by even more on both axes at once. Medverse’s own coarse-to-fine mechanism is separately prior-sensitive: feeding its own imperfect coarse prediction forward as a query prior hurts its harness cascade universally, across all 7 OOD sources tested, sometimes catastrophically (MSD Hippocampus  $0.691 \rightarrow 0.050$ ) – compounding coarse-level error appears to be a property generic to coarse-to-fine cascades, not specific to either architecture’s design. TotalSeg CT and TotalSeg MRI are in-distribution modalities for our model, which trains on both; Medverse’s released weights were trained on [...]. The gap on these two rows therefore partly reflects training data rather than architecture alone.

**A real predicted prior beats a perturbed ground-truth prior.** Same checkpoint chain, same seven validation datasets as Table 5 plus in-distribution TotalSeg CT (Table 6):

Feeding the model’s own real previous prediction forward beats a perturbed ground-truth prior on 99 of 117 classes (median  $\Delta + 0.043$ ) – a *prior-source* ablation (real vs. noisy synthetic), distinct from the Medverse *prior-presence* finding above; a true no-prior arm is still needed (gap G6). Out-of-distribution (9 held-out sources including FLARE22/NasalSeg): the prior-source gain helps 5/9 sources and hurts 4/9 – weaker than the clear in-distribution majority.

Add a no-query-prior control arm to complete the

**Table 7:** OOD Dice, real-host canvas progression. Stage 1→2 is a single-variable comparison (canvas only); stage 2→3 additionally extends to all shape families, a wider real-subject pool, and a longer budget.

| Stage                        | ISLES22      | Shifts-MS    | ATLAS v2.0 | GNC_705      |
|------------------------------|--------------|--------------|------------|--------------|
| Baseline                     | 0.050        | 0.008        | 0.028      | 0.082        |
| 1) Synthetic-bank canvas     | 0.045        | 0.011        | 0.033      | 0.070        |
| 2) Real canvas, one family   | <b>0.098</b> | <b>0.058</b> | 0.032      | 0.105        |
| 3) Real canvas, all families | 0.087        | 0.051        | 0.027      | <b>0.223</b> |

**Table 8:** TotalSegmentator validation Dice for the same checkpoints. seen/unseen: macro Dice by trained-class membership. MRI/CT: Dice by target modality.

| Stage                        | Dice         | seen         | unseen       | MRI          | CT           |
|------------------------------|--------------|--------------|--------------|--------------|--------------|
| Baseline                     | 0.418        | 0.489        | 0.344        | 0.390        | 0.468        |
| 1) Synthetic-bank canvas     | 0.422        | 0.485        | 0.356        | 0.385        | 0.470        |
| 2) Real canvas, one family   | 0.417        | 0.483        | 0.348        | 0.384        | 0.467        |
| 3) Real canvas, all families | <b>0.490</b> | <b>0.579</b> | <b>0.397</b> | <b>0.474</b> | <b>0.534</b> |

## 4.4 Synthetic Task Generation

On four held-out sources with no anatomical overlap with training (ISLES22, Shifts-MS, ATLAS v2.0, GNC\_705 – lesion targets on MRI, one Dixon), the baseline checkpoint scores Dice 0.008–0.082. Widening the shape generator’s family mix or dosage on the synthetic-bank canvas (subsection 3.4) left every source within noise of baseline. The bank turned out to be built entirely from CT scans, while every target source here is MRI – a canvas-modality mismatch, not a shape-diversity problem.

**Swapping the canvas closes the gap.** A matched pair isolates the fix: identical checkpoint, shape family (scatter\_field), dose, and training budget, only the canvas source changes from the synthetic bank to a real MRI training case from TotalSegmentator. Every source improves except ATLAS v2.0, a single compact chronic lesion – the only non-multi-focal target of the four, plausibly outside what a multiplicity-targeted family like scatter\_field can reach.

Extending the real-host canvas from one shape family to all seven, a wider real-subject pool (every organ, CT and MRI), and a longer training budget keeps compounding on GNC\_705 – the best result of the whole progression – while easing back slightly on the two brain-lesion sources from stage 2’s peak.

**No in-distribution cost.** The same checkpoints’ own periodic TotalSegmentator validation shows the OOD gain is additive, not traded against in-distribution accuracy:

**Table 9:** OOD Dice under cascade inference. 146: cascade baseline, synthetic-bank canvas. 168: 146 resumed with the real-host canvas (stage 3 above). Medverse: released weights, depth-matched native autoregressive inference (subsection 4.3); omitted for four sources where both models score too low to be informative.

| Dataset         | 146          | 168 (real-host) | Medverse     |
|-----------------|--------------|-----------------|--------------|
| HU_LWK1         | <b>0.152</b> | 0.124           | 0.020        |
| ISLES22         | 0.069        | <b>0.070</b>    | —            |
| Shifts-MS       | <b>0.031</b> | 0.019           | —            |
| ATLAS v2.0      | 0.026        | <b>0.032</b>    | —            |
| MSD Prostate    | <b>0.375</b> | 0.372           | 0.357        |
| MSD Hippocampus | 0.303        | 0.518           | <b>0.686</b> |
| GNC_705         | 0.024        | <b>0.035</b>    | —            |
| FLARE22         | <b>0.751</b> | 0.743           | 0.359        |
| NasalSeg        | 0.547        | 0.644           | <b>0.737</b> |

The seen/unseen and MRI/CT gaps hold at roughly the same size at every stage; stage 3’s longer budget lifts all five numbers together rather than trading one for another.

**The gain survives cascading.** Resuming the stage-3 checkpoint into coarse-to-fine cascade training (subsection 3.3) and re-evaluating the same nine OOD sources tests whether the real-host canvas helps under a different training and inference regime, not only the single-level one it was found under.

The real-host canvas still wins or ties on five of nine sources under cascade inference, including a 0.21 gain on MSD Hippocampus, without closing the gap to Medverse on the two sources where Medverse’s full-FOV pyramid already dominates.

Stage 2→3 changes several axes at once (family count, host pool, training budget); only stage 1→2 is a controlled ablation. N-1

## **5 Discussion**

## 6 Conclusion

### ToDoS

Select the correct language in the first rows of `main.tex`. The % makes a line a comment.

Generate a PDF (Visual Studio Code: click on the green arrow in the top right corner).

Check the PDF (Visual Studio Code: click on the two rectangles with the magnifying glass in the top right corner).

Talk about this template with your supervisor: Should you add a declaration for the use of AI? Currently no official AI-declaration is given. You can find one in the pull requests of the github repository.

With CTRL pressed you can click on the PDF and jump to the corresponding place in your `.tex` files.

Check the Errors. (Visual Studio Code: bottom left) There should be one warning because the first ToDo pushes down the second one: LaTeX Marginpar on page 2 moved.

In the `main.tex` you can press ALT + SHIFT and the ARROW UP and DOWN keys to create a multiline cursor, comment out all the sections you are currently not working on and remove the multiline cursor with ESC.

Keep the following examples and todos till the end and check them out sometimes.

Adjust your License Notes to reflect the used programs.

For Printing: Remove the `todonotes`-package in `setup.tex` to find all Todos.

**Table 10:** A nice table with a nice caption.

| Value & Unit           | Value | Unit              |
|------------------------|-------|-------------------|
| 9.82 m s <sup>-2</sup> | 9.82  | m s <sup>-2</sup> |

For Printing: Choose the correct formatting (search `print` in `setup.tex`).

You can remove the glossary and the symbolslist to make it easier, especially in a Bachelor's Thesis.

## 6.1 Tables and References

Table 10

This is a cited book [3].

Store and display plots and other graphs as `.pdf` files.

Include Algorithm 1 in the same way you include equations like Equation 4:

$$\begin{aligned}y &= 3x^2 + 5x^2 + 4x + 4x + 7 + 1 \\&= 8x^2 + 8x + 8 \\y &= 8 \cdot (x^2 + x + 1)\end{aligned}\tag{4}$$

## 6.2 Auto-complete in VSCode

There are also some auto-complete features: type `@` + letter to get the letters greek version.

Type `@v` + letter for the alternative form (like  $\varphi$  for  $\phi$ ).

You can also use shortcuts for environments like

$$BAL\tag{5}$$

Type BAL and select the correct option from the autocomplete. Maybe you have to try various ones. The B is the base command and AL is the abbreviation for align.

If you want an environment without an index, you can use BS as base command.

Same for changes to your font, like bold or italic: FIT gives *text*

## 6.3 ", Quotations and Code

Use " or file-names, file\_names and code\_snippets.

**Algorithm 1:** The Colors from the Corporate Design of the University of Freiburg. The keywords are bold and the comments blue. This can be changed in setup.tex.

```
from dataclasses import dataclass
```

```
# https://cd.uni-freiburg.de/farben/
```

```
@dataclass
```

```
class MainColors:
```

```
    blue = "#344A9A"
```

```
    black = "#000000"
```

```
    white = "#FFFFFF"
```

```
@dataclass
```

```
class BackgroundColors:
```

```
    blue_100_percent = "#344A9A"
```

```
    blue_80_percent = "#5D6BAD"
```

```
    blue_60_percent = "#868DC2"
```

```
    blue_80_percent = "#AFB1D8"
```

```
    blue_20_percent = "#D7D8EC"
```

```
    dark_blue = "#00004a"
```

```
    sand_100_percent = "#f6f1e3"
```

```
    sand_50_percent = "#faf8f1"
```

```
@dataclass
```

```
class AdditionalColors:
```

```
    green_100_percent = "#00a082"
```

```
    green_80_percent = "#27b29b"
```

```
    green_60_percent = "#7bc6b4"
```

```
green_80_percent = "#afdace"  
green_20_percent = "#daede7"  
brown_100_percent = "#8f6b30"  
brown_80_percent = "#a58959"  
brown_60_percent = "#bca683"  
brown_80_percent = "#d2c4ac"  
brown_20_percent = "#e9e1d6"  
yellow_100_percent = "#ffe863"  
yellow_80_percent = "#ffed82"  
yellow_60_percent = "#fff1a1"  
yellow_80_percent = "#fff6c1"  
yellow_20_percent = "#fffae0"  
pink_100_percent = "#f5c2ed"  
pink_80_percent = "#f7cef1"  
pink_60_percent = "#f9daf4"  
pink_80_percent = "#fbe7f8"  
pink_20_percent = "#fdf3fb"
```

## 6.4 Clipboard Manager, Git and more

Clipboard Manager help with copying multiple different things at once. On Windows you can press WINDOWS + V to activate the history and choose older things you copied.

Use Git ([click here](#)).

## 6.5 Glossary

Click them, the page numbers in the glossaries are also clickable.

name one

name one

name one

long form of name two (short2)

short2

short2

*a symbol*

*a symbol*

*a symbol*

Big Letter at start: Name one

plural of name one

name one

Big Letter at start: Short2

short2

short2s

Big Letter at start: *A symbol*

*a symbol*

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- 1 The Colors from the Corporate Design of the University of Freiburg. The keywords are bold and the comments blue. This can be changed in `setup.tex`. . . . . 24

## Glossary

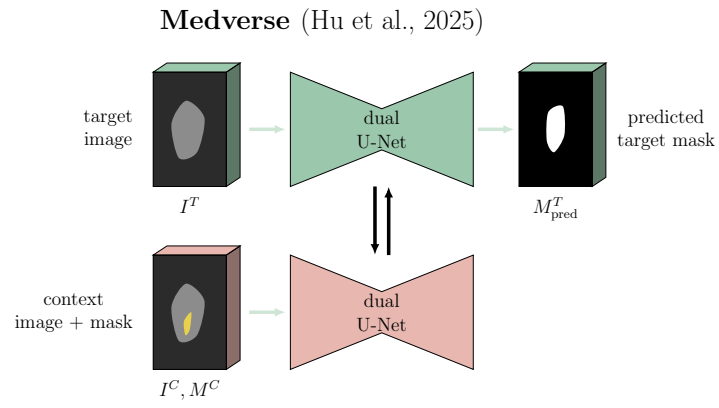
| Notation | Description     | Page List |
|----------|-----------------|-----------|
| name one | description one | 21, 23    |
| short2   | description two | 21–23     |

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# Appendix



**Figure 5:** Medverse’s architecture: dual U-Net cross-attention, shown for comparison against our architecture (Figure 1).